Candidate Number

Candidate Name____________________________

INTERNATIONAL ENGLISH LANGUAGE TESTING SYSTEM

Academic Reading

Additional materials:

Answer sheet for Listening and Reading
Time 1 hour

INSTRUCTIONS TO CANDIDATES

Do not open this question paper until you are told to do so.
Write your name and candidate number in the spaces at the top of this page. Read the instructions for each part of the paper carefully.
Answer all the questions.
Write your answers on the answer sheet. Use a pencil.
You must complete the answer sheet within the time limit.
At the end of the test, hand in both this question paper and your answer sheet.

INFORMATION FOR CANDIDATES

There are **40** questions on this question paper.
Each question carries one mark.

READING PASSAGE 1

*You should spend about 20 minutes on Questions **1–13**, which are based on Reading Passage 1 below.*

Secrets of the Nasca Lines

New research sheds light on the mysterious desert drawings of Peru

In South America, high in the Nasca desert of Peru, there are hundreds of ancient lines drawn on the ground. Some of these drawings, or geoglyphs as they are known, measure hundreds of meters across, and depict animals, birds and geometric designs. At one time or another, the Nasca lines have been explained as roads, irrigation plans, images to be appreciated from primitive hot air balloons, and even — most bizarrely — landing strips for alien spacecraft.

In the 1940s, a German-born teacher named Maria Reiche made the first formal surveys of the Nasca geoglyphs. Although her own preferred theory — that the lines represented features of an astronomical calendar — has been largely discredited, Reiche played a critically important role in conserving the geoglyphs for a half-century. Then in 1997, the Nasca-Palpa Project, a Peruvian-German research collaboration, was launched. Directed by Peruvian Johny Isla, and Markus Reindel of the German Archaeological Institute, the project has mounted a systematic study of the ancient people of the region, starting with where and how the ancient Nasca people lived, why they disappeared, and what was the meaning of the strange designs they left behind. Isla and Reindel believe that the story begins, and ends, with water.

The coastal region of southern Peru and northern Chile is one of the driest places on Earth. In the small, protected area where the Nasca culture arose, ten rivers descend westwards from the Andes mountains, most of them dry at least part of the year. Despite these difficult conditions, the Nasca flourished for eight centuries. Around 200 BC, the Nasca people emerged out of a previous culture known as the Paracas, settling along the valleys and cultivating crops such as cotton, beans, *lucuma* fruit, and a short eared form of corn. Renowned for their distinctive pottery, they invented a new technique of mixing natural mineral pigments in a thin wash of clay so that colors could be baked into it.

The Nasca settlements moved east or west as local annual patterns of rainfall alternated. Early Nasca engineers may have also designed a more practical approach to coping with the water shortage. An ingenious system of wells, tapping into the sloping water table as it falls from the Andean foothills, allowed settlements to bring subterranean water to the surface. Perhaps due to the adversity they faced, the Nasca people seem to have led a remarkably green life. They appear to have had a complex sense of water conservation since the underground water system could minimize evaporation. The layers of vegetative matter in the walls of buildings and terraces suggest that they even recycled their rubbish to use it as a building substance.
And the geoglyphs? Who first created them? The parched desert and hillsides made an inviting canvas: by merely clearing away a layer of stones cluttering the ground and exposing the paler sand below, the Nasca gave birth to markings that have lasted for centuries in the dry climate. But the story does not begin there. On a hillside south of the town of Palpa are drawings of three stylized human figures that date back at least 2,400 years — to the era of the earlier Paracas civilization.

Recent examinations of the line drawings of the area indicate that they were not produced at one time, in one place, to achieve one goal. Many have been superimposed on older ones, with erasures and overwriting, making their interpretation increasingly more complicated. Almost all of the iconic animal figures, such as the spider and the hummingbird, were single-line drawings. This means a person could step into them at one point and exit at another without ever crossing a line, suggesting that at some point in early Nasca times, the lines evolved from simple images to pathways for processions in ceremonies. Later, potentially as a response to population growth, more people may have participated in these rituals, and the geoglyphs took on open, geometrical forms and shapes, with some trapezoid shapes stretching longer than 700 meters. "Our idea," Reindel says, "is that they were no longer meant as images to be seen, but became stages to be walked upon, to be used for religious ceremonies."

Those ancient ways of worship have left their traces in the ground itself. In the four years following 2003, German geophysicists measured the Earth's magnetic field on a trapezoid near Yunama, a village close to Palpa, and on other lines nearby. Slight changes in the magnetic signal indicated that human activity compacted the sand, especially around the piles of stones located at the ends of many of the drawings.

Archaeologists have long suspected that these stone piles were altars for ceremonies. In 2000, their ideas were eventually confirmed. As Reindel dug his way through one pile near Yunama, he uncovered relics that evidently represented ritualistic offerings, along with fragments of a large seashell of the genus *Spondylus*, distinctive for its creamy, coral-like hues and spiky outside. The *Spondylus* shell was a very important religious symbol for agricultural fertility and is connected in certain activities, as prayer for relief from drought. And it's clear, Reindel says, "In this area, water was the key issue." Ultimately, all those offerings and prayers were ignored. By the 6th century AD, the desert grew larger, and the Nasca civilization collapsed, leaving its ancient secrets carved in its dryness.

Questions 1–7

Do the following statements agree with the information given in Reading Passage 1? In boxes 1–7 on your answer sheet, write:

TRUE *if the statement agrees with the information*

FALSE *if the statement contradicts the information*

NOT GIVEN *if there is no information on this*

1. There have been several theories on the purpose of the Nasca geoglyphs.
2. Today, most people still agree with Maria Reiche on the geoglyphs.
3. The Nasca-Palpa Project is the most thorough survey of the geoglyphs carried out so far.
4. The Nasca were the first people to produce geoglyphs in the desert of Peru.
5. The shapes of the geoglyphs had been constant throughout the Nasca period.
6. Scientific analysis backs the idea that some geoglyphs were made for walking.
7. The *Spondylus* shell indicates a link between the geoglyphs and the essentiality of water.

Questions 8–13

Complete the notes below.
Choose ***ONE WORD ONLY*** *from the passage for each answer.*
Write your answers in boxes 8–13 on your answer sheet.

The Nasca people

• grew from a previous culture known as the Paracas.

• lived in settlements in the valleys in close vicinity to the Andes mountains.

• grew food crops and cotton.

• created **8** __________ with multiple colors by using a wash of clay and pigments.

• moved their settlements following changes in annual **9** __________.

• built a network of **10** __________ to gain access to underground water.

• had a 'green lifestyle', for example, used **11** __________ as construction materials.

• made drawings on the ground by removing **12** __________ to reveal lighter colored **13** __________ below.

READING PASSAGE 2

*You should spend about 20 minutes on Questions **14–26**, which are based on Reading Passage 2 on the following pages.*

Questions 14–18

*Reading Passage 2 has five sections, **A–E.***
Choose the correct heading for each section from the list of headings below.
*Write the correct number, **i–viii**, in boxes **14–18** on your answer sheet.*

List of Headings
i The need for skill and care
ii Choosing the richest asteroid
iii The safest way to protect an asteroid
iv Obtaining support for the project
v An achievable goal, not an impossible dream
vi The need for global cooperation
vii Beginning with a less challenging task
viii Practical, economic and research justifications

14 Section **A**
15 Section **B**
16 Section **C**
17 Section **D**
18 Section **E**

The plan to bring an asteroid to Earth

Moving in orbit around our Sun are millions of rocks known as asteroids. Now scientists have plans to capture one.

A Send a robot into space, catch an asteroid and bring it back to Earth's orbit. This, say scientists and engineers at the California Institute of Technology (Caltech), could be feasible. A four-day workshop was dedicated to investigating the feasibility of capturing a near-Earth asteroid, bringing it closer to our planet and using it as a base for future manned space-flight missions.

This is not something the scientists are imagining could be done someday in the future. It is possible with the technology we have today and could be accomplished within a decade. A robotic probe could anchor to an asteroid with simple magnets or grab it with specialised claws. Alternatively, a large spacecraft could use its gravitational field to shift the orbit of a larger asteroid, sending it towards Earth.

'Once you get over the initial reaction — *You want to do what?!* — it actually starts to seem like a reasonable idea,' says engineer John Brophy, who helped organise the workshop. In fact, many of these ideas have been on the drawing board for years as part of NASA's planetary defence programme against large space-based objects that might threaten Earth. And there's no shortage of potential targets: NASA estimates there are 19 500 asteroids at least 100 metres wide within 45 million kilometres of Earth.

B Though rearranging the heavens may seem an excessive undertaking, this US mission has its merits. The US already has plans to send astronauts to a near-Earth asteroid, a mission that would mean confining them in a tiny capsule for three to six months and would involve all the risks of a deep-space voyage. Instead, robots could bring an asteroid close enough for them to get there in just a month.

An asteroid would provide a stationary base from which to launch missions further into space. There are several advantages to this. For one, launching missions carrying materials from Earth requires a lot of power, fuel and, consequently, money, because of our planet's strong force of gravity. Since this would be far weaker on an asteroid, materials mined there could be more easily taken off the asteroid and shuttled around the solar system.

Many asteroids have a lot to offer. Some are rich in metals, which can be mined and used to build space-based habitats, or brought back to Earth. Others are up to one-quarter water, which could be used for life-support or broken down into hydrogen and oxygen to make fuel. Astronomers would have the chance to get a close-up look at one of the solar

system's earliest relics, generating important scientific data. 'Executing the asteroid-retrieval plan would help demonstrate and greatly expand mankind's space-based engineering capabilities,' says engineer Louis Friedman, another co-organiser of the Caltech workshop. For instance, the mission would
teach engineers how to capture an unco-operative target, which could be useful practice for planetary-defence missions in the event of a threat from a meteoroid or comet approaching our planet,' he adds.

C Former astronaut Rusty Schweickart, co-founder of the B612 Foundation, an organisation dedicated to protecting Earth from asteroid strikes, points out that although it would be technically feasible, shifting such a hefty and substantial target would not be easy: 'You're moving the largest mother-lode imaginable,' he says.

Engineers would need to be absolutely certain they could control such a potentially dangerous object. 'It's the opposite of planetary defence; if you do something wrong you have a Tunguska event,' says engineer Marco Tantardini from the Planetary Society, referring to the powerful 1908 explosion above a remote Russian region thought to have been caused by a meteoroid or comet.

D Still, these obstacles only add to the appeal of the project for engineers, who love to go over every potential difficulty in order to solve it. And if the challenges posed by a large asteroid seem too daunting, researchers could always start with a smaller asteroid, perhaps two to ten metres across. Last year, Brophy helped conduct a study to look at the feasibility of bringing a two-metre, 1 000-kilogram asteroid — of which there might conceivably be millions — to the International Space Station. The study suggested the asteroid could be captured robotically in something as simple as a large bag. Of course, such a small object might not have the same emotional impact as a larger target. 'NASA isn't going to want to go to something that is smaller than our spaceships,' says engineer Dan Mazanek from NASA's Langley Research Center.

E No matter the size of the asteroid, these plans would require hefty investments. Even capturing a small asteroid would consume at least a billion dollars. Convincing taxpayers to foot such a bill could be difficult. However, private industry might be interested in getting involved. One possibility would be to push the asteroid into near-Earth orbit and then invite anyone who wants to develop the capabilities to reach and mine the object.

However, although the undertaking might be scientifically exciting and provide great insight into the solar system's formation, this is not enough on its own to justify the expense of bringing an asteroid to Earth. Investigations of asteroids can be done much more cheaply with an unmanned spacecraft, says chemist Joseph A. Nuth from NASA's Goddard Space Flight Center. According to Brophy, ultimately we would be working towards bringing an asteroid closer to Earth in order to help humanity move out into the solar system.

Questions 19 – 22

*Look at the following experts (Questions 19 – 22) and the list of statements below. Match each expert with the correct statement, **A–G**.*
*Write the correct letter, **A–G**, in boxes 19 – 22 on your answer sheet.*

19. Louis Friedman
20. Rusty Schweickart
21. Dan Mazanek
22. Joseph A. Nuth

List of statements

A. A mistake could have serious consequences for Earth.

B. It might be difficult to arouse interest in an asteroid of limited size.

C. The project could be an early step in space exploration.

D. An asteroid's weight makes the project extremely challenging.

E. The skill gained could save Earth from future danger.

F. An asteroid could be a useful landing place for a spaceship.

G. Capturing an asteroid would not be an efficient method of research.

Questions 23 – 26

Complete the summary below.
Choose ***ONE WORD ONLY*** from the passage for each answer.
Write your answers in boxes 23 – 26 on your answer sheet.

The merits of the US mission

Capture of an asteroid would reduce the time that astronauts travelling to it needed to spend in space. An asteroid could also act as a **23** ______ for further space travel and exploration. The fact that an asteroid would have weaker **24** ______ would allow easier movement of resources.
These resources include **25** ______ which could be used in space or on Earth, and **26** ______ which could be consumed or used as a source of power.
The mission could provide data on the history of our Sun and planets. It could also be good practice if there was a threat to Earth from a body from space.

READING PASSAGE 3

You should spend about 20 minutes on Questions 27–40, which are based on Reading Passage 3 on pages 10 and 11.

Sir Francis Ronalds 1788–1873

An early pioneer in the story of long-distance communication

A Francis Ronalds was born in London in 1788, the second of eleven children. When he was nineteen his father died, and as the eldest son, the young Ronalds had to assume the responsibility of running the family's cheese business. He combined this occupation with his passions for chemical experiments and for meteorology, the study of the Earth's atmosphere and its changes. This brought him into contact with the elderly Swiss meteorologist Jean André de Luc, who was also an inspired inventor in the field of electricity. In 1814, de Luc encouraged Ronalds to start work on this new discipline.

B For the next few years, Ronalds experimented enthusiastically with various kinds of electrostatic clockwork devices. He was particularly interested in the idea of electric telegraphy: transmitting messages over distance, using electrical impulses carried along metal wires. In the best traditions of nineteenth-century scientists, these investigations were carried out at home. In early 1816, Ronalds gutted the family home in Hammersmith, West London, and covered its rooms, halls, and staircases with electric wire. This experiment involved the use of current to produce movement in small corks at the ends of circuits.

C Some months later, Ronalds built two large wooden frames accommodating 8 miles (12.9 km) of electrically charged iron wire in his back garden. At either end of the wire was a revolving disc, on which letters and numbers were engraved. These dials were kept revolving by means of a clockwork mechanism. When somebody wanted to send a message, he would wait until the dial at his end was showing the desired number or letter, and then interrupt the charge on the wire. This would cause two balls suspended on the wire to fall together. The person at the opposite end would then note down what was indicated by the dial at his end at the moment when the balls made contact. Ronalds soon developed a modified version of his invention in which the wire was encased in glass tubes and buried in the ground. The system was slow and depended on the two dials staying in step, but it worked.

D Ronalds immediately contacted the Admiralty, the department of the British government which had control over the nation's naval affairs. He was not the first to propose a system of this nature to them. Some ten years earlier, in 1806, the inventor Ralph Wedgwood had offered the Admiralty a similar type of telegraph based on frictional electricity, which it had rejected. The Admiralty had its home in London, whereas its naval fleets were based over 130 km away in Portsmouth. With communication between these two cities in mind, Ronalds wrote to Lord Melville, the then First Lord of the Admiralty, to offer him his telegraphic system.

E Ronalds believed that his invention was capable of carrying messages some 800 km, and in his letter to Lord Melville written on 11 July 1816 he spoke of the 'mode of conveying telegraphic intelligence with great rapidity, accuracy, and certainty, in all states of the atmosphere, either at night or in the day, and at small expense.' In 1816, the Admiralty had at its disposal only the late eighteenth-century telegraphic system of semaphore, whereby a mechanical device was used to control a mast with two arms. The disadvantage of this system was that it was only usable in daylight and in clear weather conditions.

Nevertheless, John Barrow, Secretary to the Admiralty, wrote back to Ronalds on 5 August, saying that 'telegraphs of any kind are wholly unnecessary; and NO other than the one now in use will be adopted.'

F Ronalds did not pursue the matter further or attempt to patent his telegraph. However, in a small pamphlet published in 1823, he described his invention and listed some of its potential uses: 'Why should not government govern at Portsmouth almost as promptly as in Downing Street? Why should our defaulters escape by default of our foggy climate? Let us have Electrical Conversation offices communicating with each other all over the kingdom if we can.' Although Ronalds lived long enough to see his prophecies come to fruition, he made NO further contribution to the progress of telegraphic communication. In 1843, he was appointed the first Honorary Director and Superintendent of the Observatory at Kew, where he devised an accurate, automatic system for registering meteorological data using photography. A similar system was developed independently by Charles Brooke, but the British Association of Meteorologists confirmed Ronalds's priority. His system remained in use for many decades.

G Meanwhile, the people who benefited most directly from Ronalds's telegraphic work were the physicist Charles Wheatstone and his business partner William Cooke. They saw the potential of a rapid communication system for the growing number of private railway companies, and in 1837 they developed the Wheatstone–Cooke five-needle telegraph. The pair patented their design in the same year, the first of its kind to be officially registered, and reaped considerable financial gain as a result. Ronalds himself retired in 1852 and devoted his later years to collecting and cataloguing a vast library of published material on electricity and magnetism. In 1870, three years before he died, he was knighted by Queen Victoria, a belated honour in recognition of his 'early and remarkable labours in telegraphic investigations.'

Questions 27–31

Look at the following events (Questions 27–31) and the list of dates below. Match each event with the correct date, ***A–H****.*
Write the correct letter, ***A–H****, in boxes 27–31 on your answer sheet.*

27. The first electric telegraph system was submitted to the Admiralty.

28. Ronalds received official recognition of his contribution to telegraphy.

29. The first electric telegraph was patented.

30. Ronalds began to experiment with electricity.

31. Ronalds informed the public about the possible uses of his telegraph.

List of Dates

A. 1806
B. 1814
C. 1816
D. 1823
E. 1837
F. 1843
G. 1852
H. 1870

Questions 32–35

Answer the questions below.
Choose ***NO MORE THAN THREE WORDS AND/OR A NUMBER*** *from the passage for each answer. Write your answers in boxes 32–35 on your answer sheet.*

32. What was marked on the dials on Ronalds's garden telegraph?

33. What did Ronalds use to enclose the wires on his later version of the garden telegraph?

34. How far did Ronalds believe his telegraph could transmit information?

35. What did the communication system invented by Ralph Wedgwood use?

Questions 36–40

Reading Passage 3 has seven paragraphs, ***A–G****.*

Which paragraph contains the following information?
Write the correct letter, ***A–G****, in boxes 36–40 on your answer sheet.*

36. a description of a commercial application for the electric telegraph
37. a rejection of Ronalds's invention
38. a drawback of Ronalds's invention
39. reference to the positive influence on Ronalds of a fellow scientist
40. reference to the adoption of one of Ronalds's procedures in preference to that of another scientist